

Shah Nurul Hasnat Sadi

Dhaka, Bangladesh. • +8801723350114 • sadi.49@geography-juniv.edu.bd

Website • [ORCiD](#) • [LinkedIn](#) • [GitHub](#) • [ResearchGate](#)

Professional Summary:

Geospatial researcher and Research Fellow at SPARRSO, specializing in automated remote sensing workflows and cloud computing. Highly proficient in Google Earth Engine and Python, actively developing multi-sensor fusion datasets and algorithm-driven environmental models. Combines strong analytical research capabilities with practical expertise in UAV and LiDAR processing to deliver robust, scalable geospatial solutions. Currently seeking fully funded PhD/MSc opportunities and advanced Research Assistant positions to further advance research in spatial modeling and automated environmental monitoring.



Academic Qualifications:

- **M.Sc. in Geography and Environment** August, 2025 - April, 2026
Department of Geography and Environment, Jahangirnagar University. CGPA: (Appeared)
Dissertation: *Percentile-Based Time-Series Compositing and SWIR-Red Indexing for Automated Intertidal Mudflat Extraction in Eastern Meghna Estuary.*

- **B.Sc. (Hons) in Geography and Environment** March, 2020 - July, 2025
Department of Geography and Environment, Jahangirnagar University. CGPA: 3.50/4.00

Major Courses: Geoinformatics, Remote Sensing, Cartography, Spatial & Environmental Analysis, Hydrology, Climatology, Descriptive Statistics, Digital Image Processing, Urban and Rural Geography, Research Methodology.

Publications:

Shah Nurul Hasnat Sadi, M. Mahmudur Rahman, Md. Alamgir Hossen Bhuiyan, Rezaul Roni, Md Saifuzzaman, Mehedi Hasan Peas. “A Novel Approach for Automated Identification of Intertidal Mudflats in Dynamic Turbid Water Conditions” (*In peer-review: International Journal of Remote Sensing, Taylor & Francis; Submitted in 1st May, 2026*)

Roni, Rezaul, Shah Nurul Hasnat Sadi, and Abdullah Al Mamun. 2025. “Quantifying aboveground carbon stock at species level using TLS LIDAR and UAV photogrammetry for urban trees.” In *Earth Sciences: Modern Geospatial Approaches for Environmental Monitoring and Management*. <https://doi.org/10.5772/intechopen.1009808>.

Professional Experience:

- **Bangladesh Space Research and Remote Sensing Organization (SPARRSO), Govt. of Bangladesh.**
- **Junior Research Fellow** October, 2025 - June, 2026

Engineered automated, cloud-based machine learning workflows in Google Earth Engine, including the implementation of an MSIC-Otsu algorithm to extract tidal mudflats in highly turbid estuaries. Additionally, developed a national-scale multi-sensor fusion dataset, integrating Sentinel, Landsat, PALSAR C-Band, VIIRS, and DEM data, to accurately map urban extents.

- **Department of Geography and Environment, Jahangirnagar University, Savar, Dhaka, Bangladesh.**
- **Training Assistant**

“Training of Trainers on Farm Management using Frontier Technology” (June 3-4th, 2025)
Organizer and Funder: National Agriculture Training Academy (NATA), Gazipur.

- **Lab Instructor**
“Training on GIS and Remote Sensing for Geography teachers of National University” (May 19-22nd, 2025)
Organizer: Dept. of Geography and Environment, JU; Funded by: Teachers Training, National University.

- **Lab Assistant**
“Training Program on Geographic Information System focusing on LiDAR” (November 7-8th, 2024)
Organizer: Wildlife Centre, Gazipur; Funded by: Sustainable Forests and Live hoods (SUFAL) Project.

Skills and Tools:

- **Geospatial:** Google Earth Engine, ArcGIS PRO, QGIS, Google Earth Pro.
- **UAV & LiDAR:** Mission planning and flight operations, 3D Survey, LasTools, TerraScan, Pix4D Mapper, CloudCompare, UgCS.
- **Programming:** Python (GeoPandas, NumPy, GeoPy, geemap, LasPy), JavaScript, JSON, Basics of SQL and MATLAB, Git & GitHub. Platforms- Google Colab, Jupyter Notebook, PyCharm, VS Code, GEE API.
- **Research and Writing:** Systematic Literature Review, Gap Finding, Report Writing, Quantitative and Qualitative Data Collection and Analysis. Softwares- Zotero, NVivo, Microsoft Office Suite & Google Workspace (G-Suite) Applications.

Projects:

- **Gender Specific Single-Age Population per Geocode of Bangladesh.** (Database) |January, 2025|[Notebook](#) 
National Population and Housing Census of Bangladesh, 2022.

Developed a Python notebook in Google Colab to process and analyze 4.8 million rows of national population census data to produce a compact gender wise single age population database for 60,420 geocodes of Bangladesh.

- **GEE APP: Flood Events Modeling with Sentinel-1 SAR: Sylhet Division.** |October, 2024 |[Application](#) 

Built an interactive Google Earth Engine (GEE) application that assessed flood extent over Sylhet division, to visualize and calculate total & portion of flooded area, utilizing the ESA Sentinel-1 SAR GRD data.

- **GEE APP: 10-class Interactive LULC-Explorer for Bangladesh.** |September, 2024 |[Application](#) 

Enabled visualization and download 10-class LULC raster tiles of max 32 MB for 8 Divisions to 5,160 Thanas in Bangladesh along with 2 infographics, utilizing the “ESA World Cover – 2020” database.

Training and Workshops:

- End-to-End Google Earth Engine (*Spatial Thoughts LLP*; ID: ST-GEE-745) |Aug 14th, 24 |[Certificate](#) 
- 7x Courses Completed on Coursera |[View-Profile](#) 
- 3x MOOC Certification from Esri Academy |[Certificates](#) 
- 8x Workshops Certification from NASA’s Applied Remote Sensing Training Program |[Certificates](#) 

Leadership and Volunteering:

- Bangladesh National Cadet Corps (BNCC) |March, 2020 - February, 2023
Cadet Corporal, Jahangirnagar University Unit (Army Wing - 3rd Ramna Regiment).

Volunteered in administrative activities, tree plantation drives, managed 6 Cadets in successful execution of the "JU Entrance Exam-2021"; Received para-military training from Bangladesh Army and achieved the rank of Cadet Corporal.

Languages:

English (Full Professional Proficiency), **Bengali** (Native), **Hindi** (Spoken Proficiency)

Hobbies:

Competitive Chess, Studying Cosmology, Playing the Guitar and E-sports.

Reference:

Rezaul Roni

MSc in Geoinformatics (University of Twente),

B.Sc. & M.Sc. (JU)

Associate Professor

Department of Geography & Environment

Jahangirnagar University

Savar, Dhaka-1342, Bangladesh

Email: georoni31@juniv.edu

Mob: +8801716049335